Perspectives on the North Branch Potomac:
Considering modern data gaps in time and space

North Branch Potomac River Environmental Flows Workshop

Elk Garden Fire House

8/6/2025-8/7/2025

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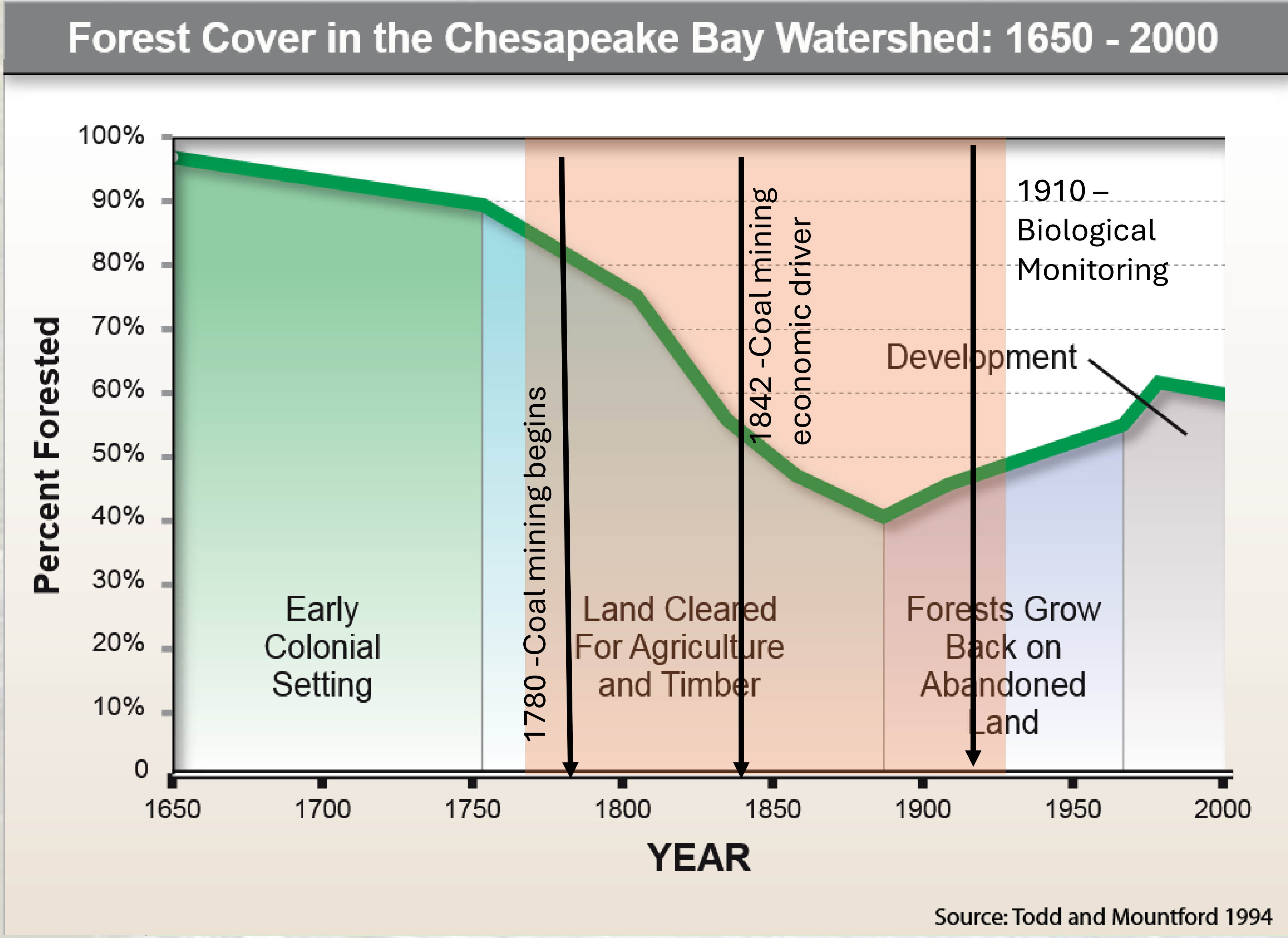
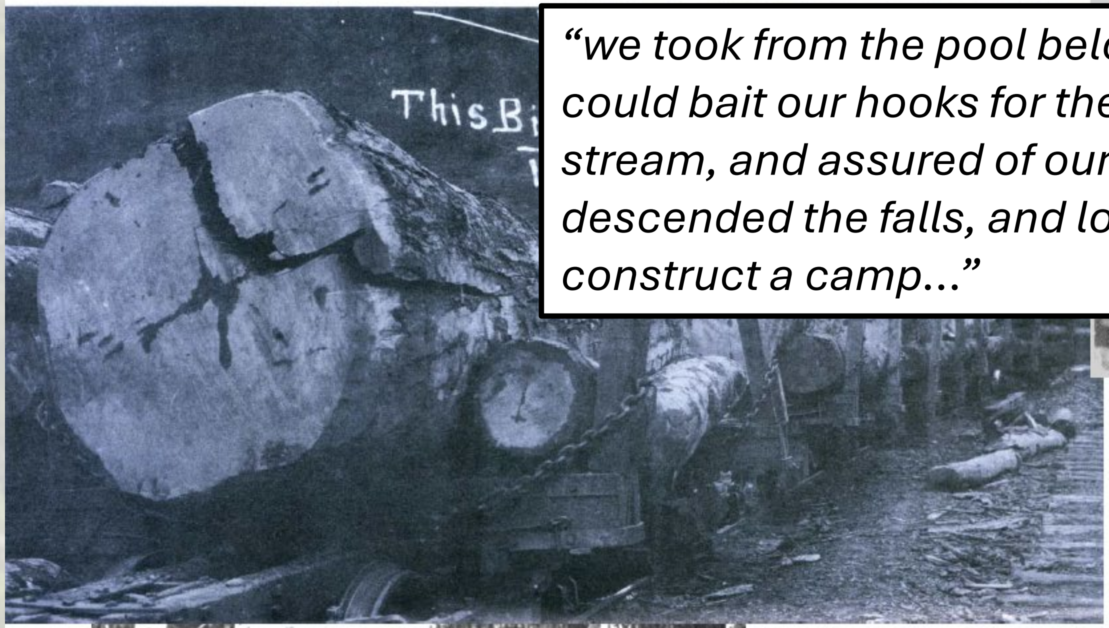
ICPRB
potomacriver.org

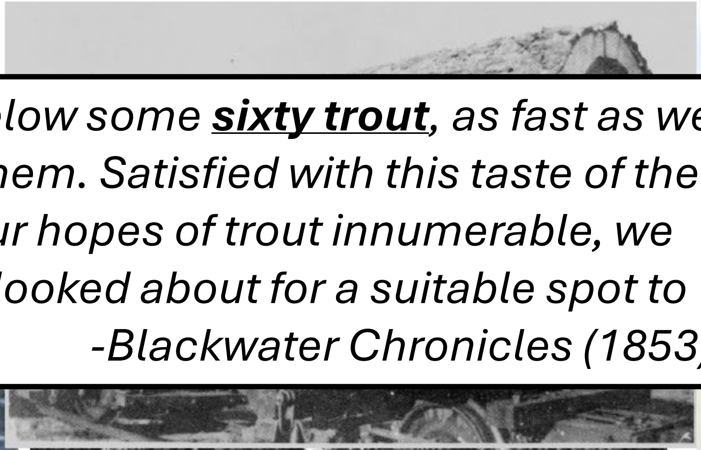
- Sediment Cores: Pollen records show a shift to agriculture (non-hardwoods) following settlement
- Early settlements were technology limited, however, large scale burns and targeting oldest most valuable trees rapidly changed dominant species



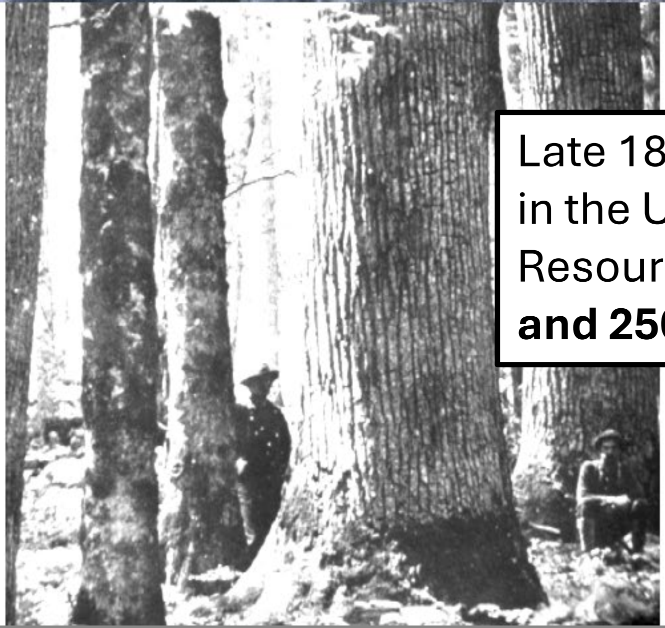
The State of Chesapeake Forests by Sprague et. al (2006)

Fig. 8—This large yellow poplar cut on Green Mountain, Tucker County, by the Otter Creek Boom and Lumber Co. at Hambleton, Tucker County, filled an entire log train and furnished 12,469 board feet of lumber, 1913. Courtesy Homer Floyd Fansler.





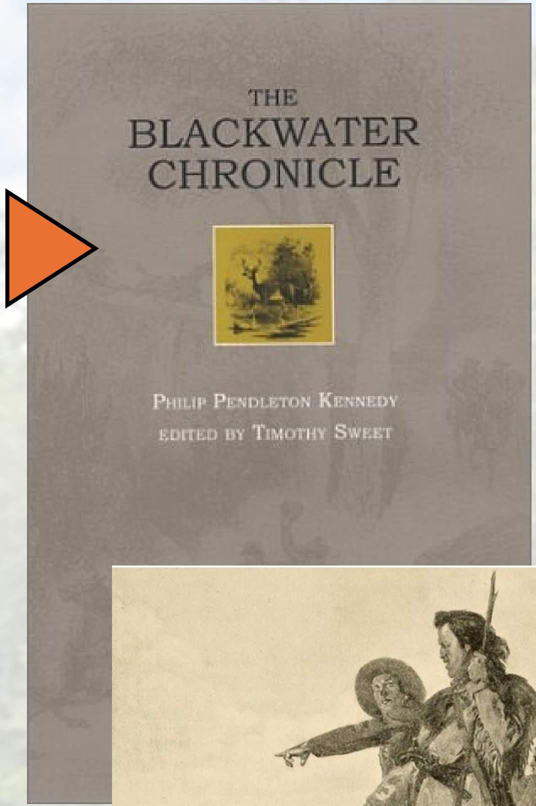
*"we took from the pool below some **<u>sixty trout</u>**, as fast as we could bait our hooks for them. Satisfied with this taste of the stream, and assured of our hopes of trout innumerable, we descended the falls, and looked about for a suitable spot to construct a camp..."* *-Blackwater Chronicles (1853)*

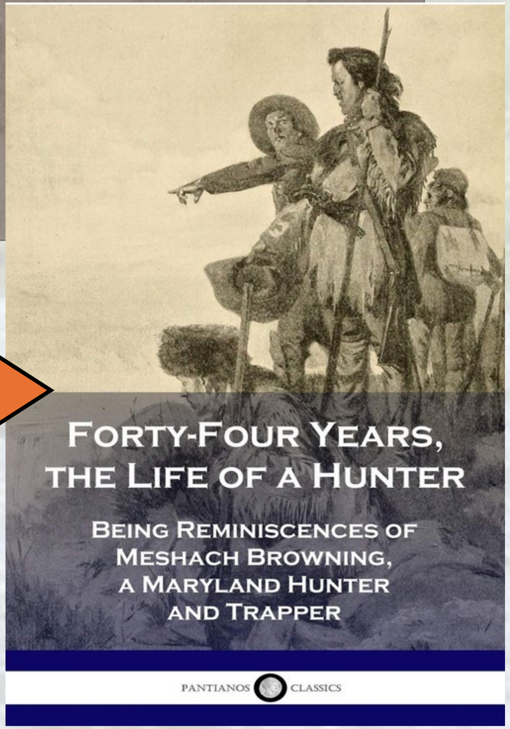


The Potomac River and the

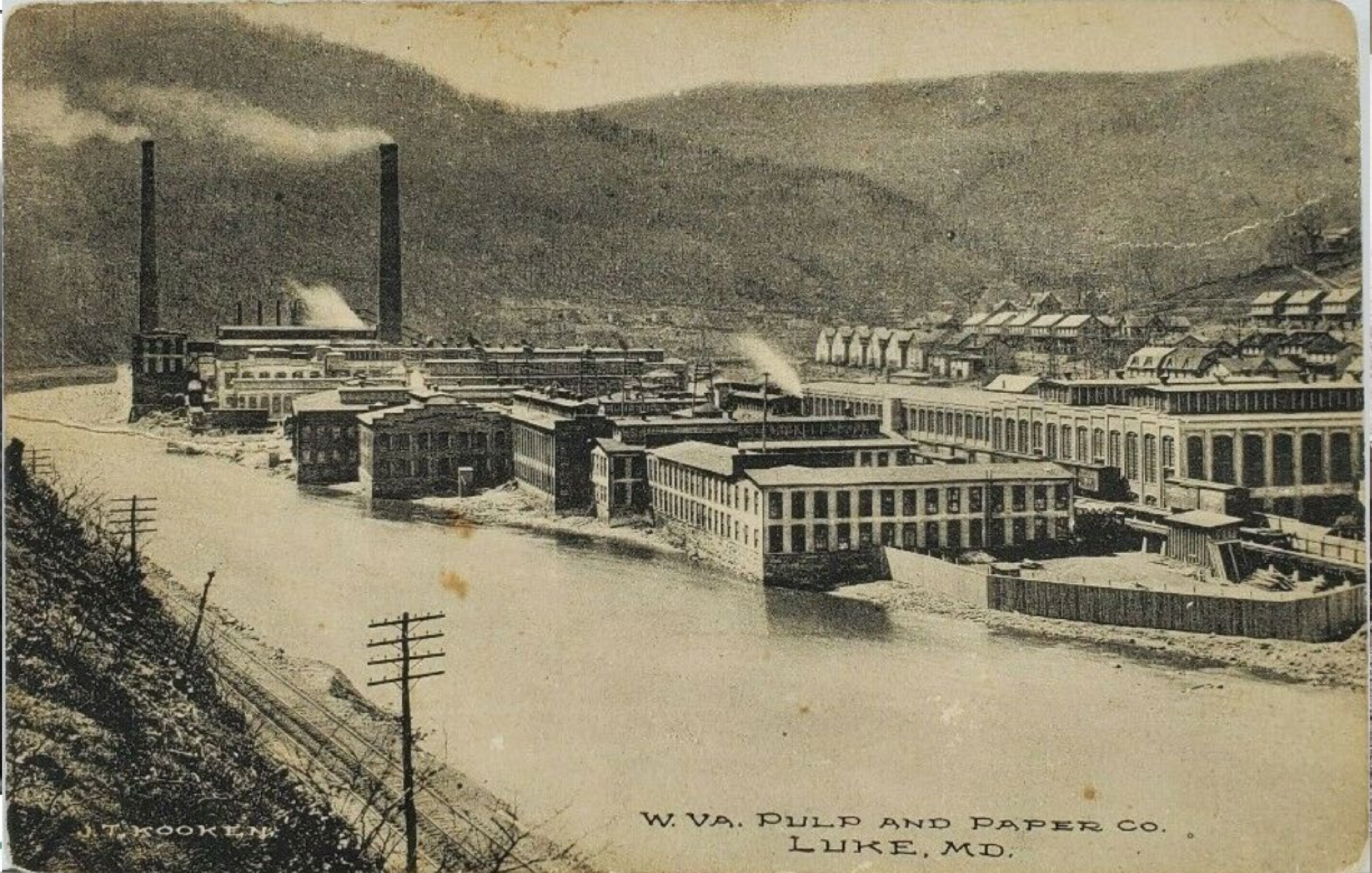
huge trees.

Late 1800's there were an estimated 300,000 whitetail deer in the USA. Currently, Maryland Department of Natural Resources estimate populations to range **between 200,000 and 250,000 in Maryland alone**.





Hillside



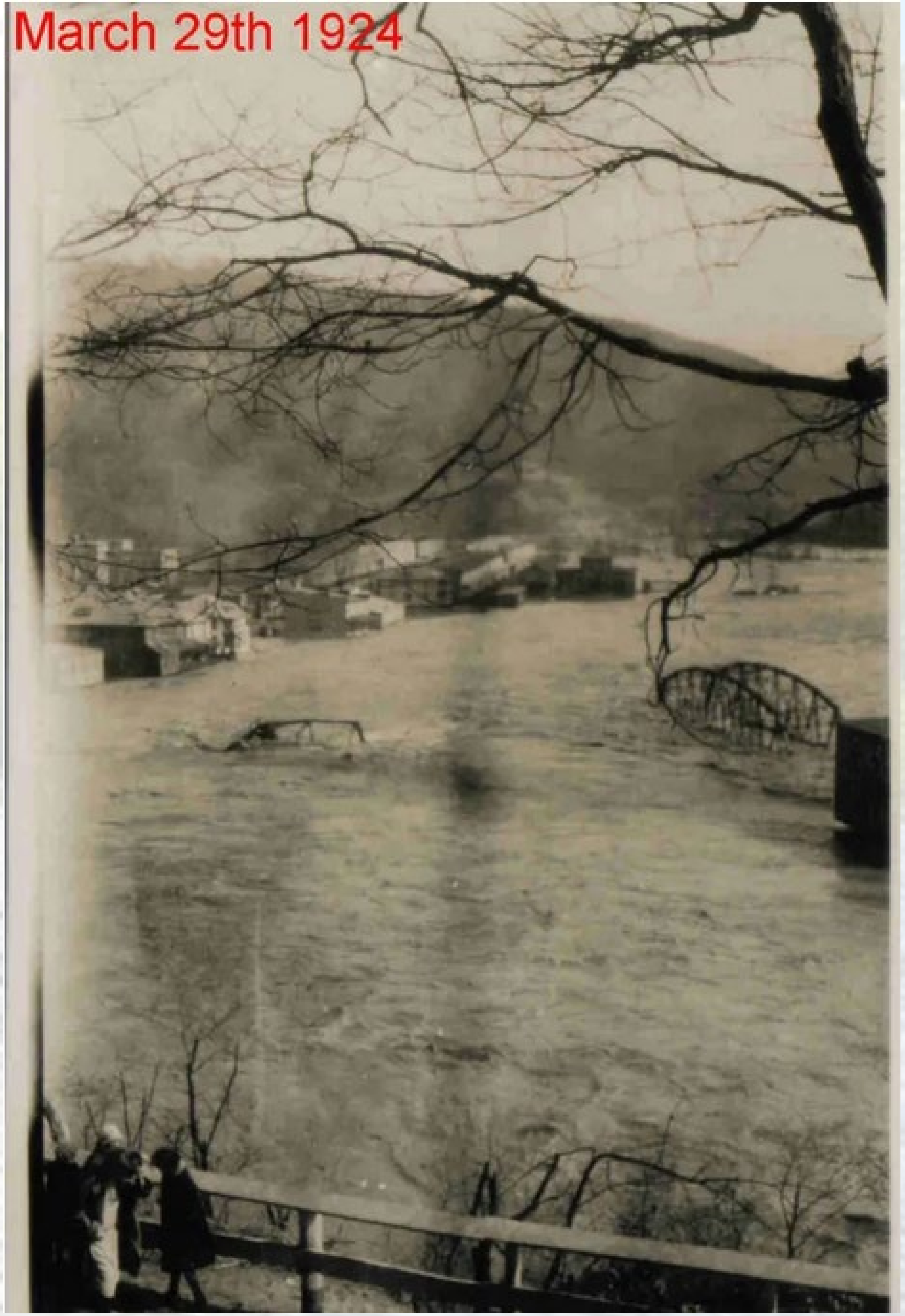
Keyser, West Virginia 1865

Keyser, West Virginia 1923

Piedmont, West Virginia 1910

"Another nor'easter struck on the 21st, this time dumping 15 to 20 inches of snow locally. The month's heavy snow left many roads blocked, and winds produced high drifts in some cases 15 to 20 feet high. There was 3 to 4 feet of snow covering the mountains. Then, on March 28, the temperature topped 70 degrees at Cumberland."

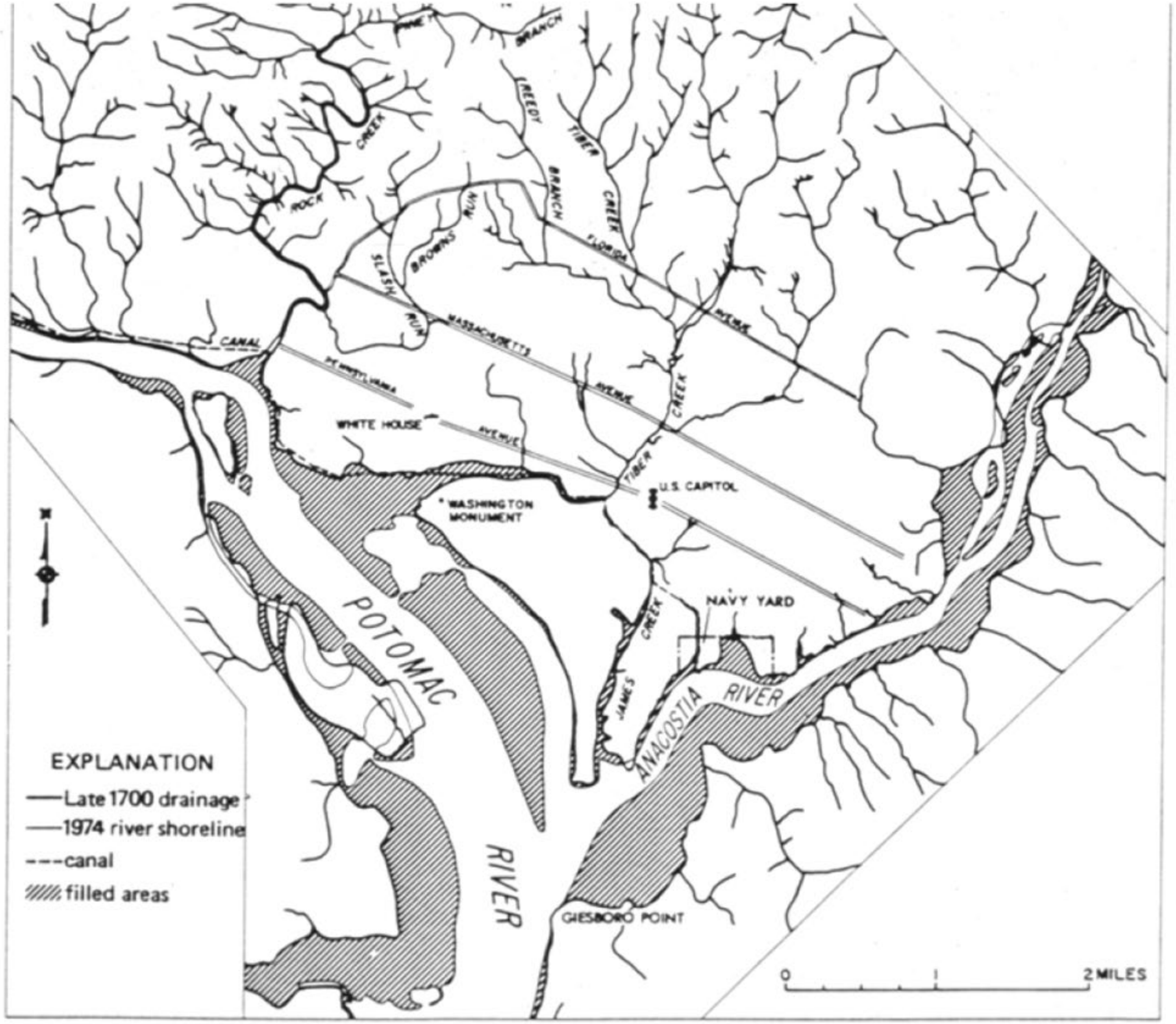
Piedmont (1924)



1770-1970: 200

Fig. 76—The barren aspect of the land following the lumbering operations and fires is starkly shown here. Crest of Cabin Mountain, near Dobbins Slashings,

Sediment settles out downstream of the fall line, just upstream from Chain Bridge.
Washington, DC is frequently mischaracterized as "built upon a swamp." The White House sits at 75' elevation and the city rises to 410' near the National Cathedral –"steep swampland". However, the shoreline of Washington has filled in tremendously over the last couple of centuries. Much of that filled-land (grey areas below), including some former wetlands, is now National Park land. Much of the new land came from bare farmlands and the denuded mountainsides of VA, WV, MD and PA.



"Soils that were s
every 600 years w

"... in 1917 alone,

The State of Chesa
by Sprague et. al (2006)

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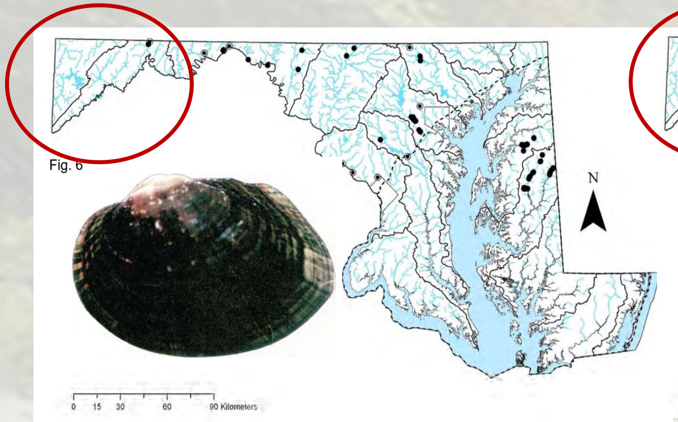
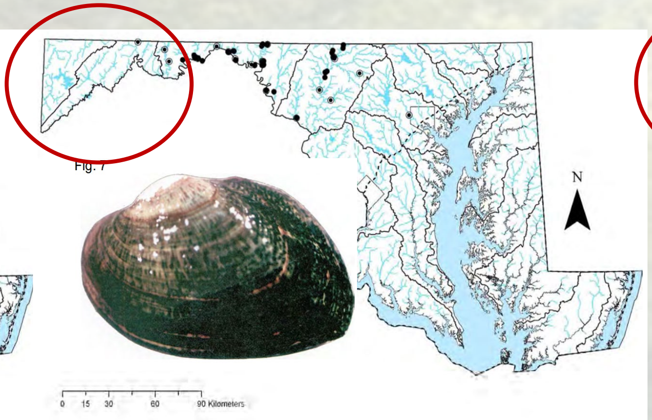
Modern vs Historical Distributions: Freshwater Mussels

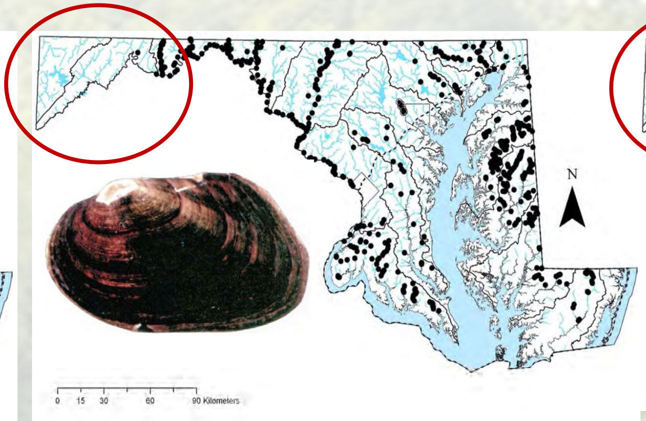
Manual of the Freshwater Bivalves of Maryland
(Bogan and Ashton, 2016)



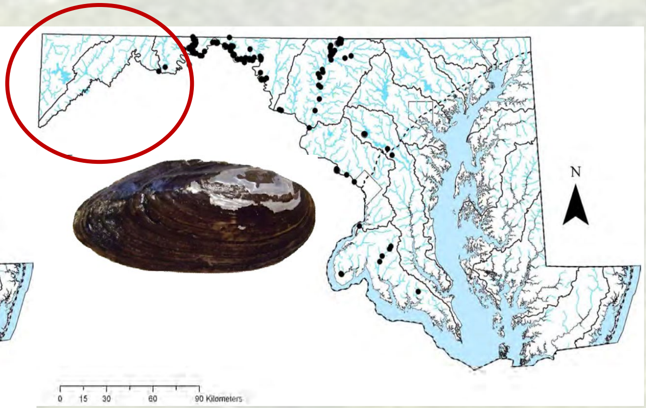
Triangle Floater (*A. undulata*)



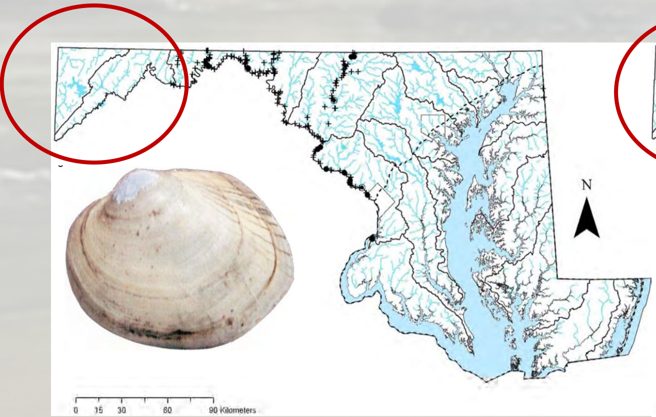
Brook Floater (*A. varicosa*)



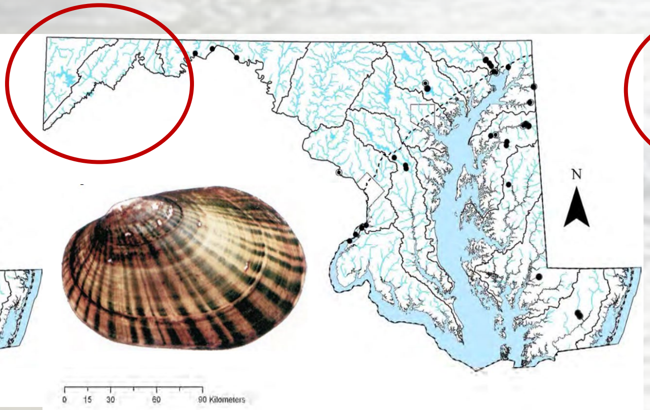
Eastern Elliptio (*E. complanata*)



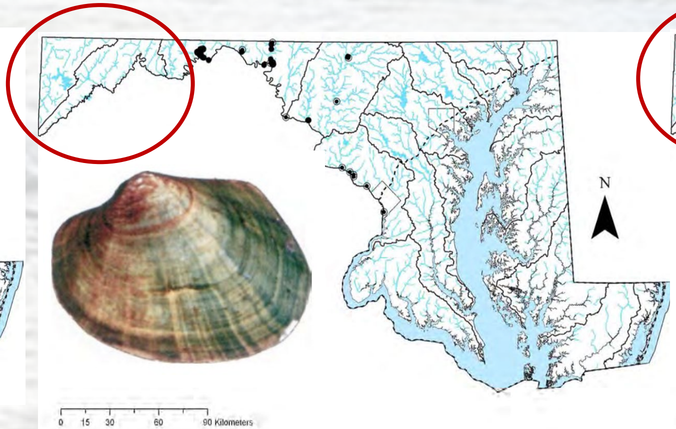
Atlantic Spike (*E. producta*)



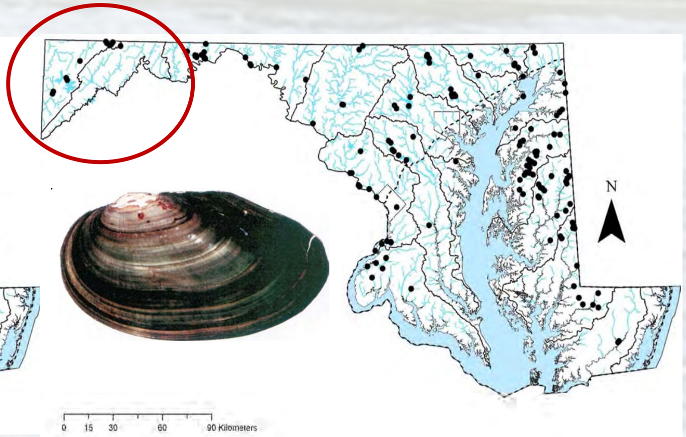
Yellow lampmussel (*L. cariosa*)



Eastern lampmussel (*L. radiata*)



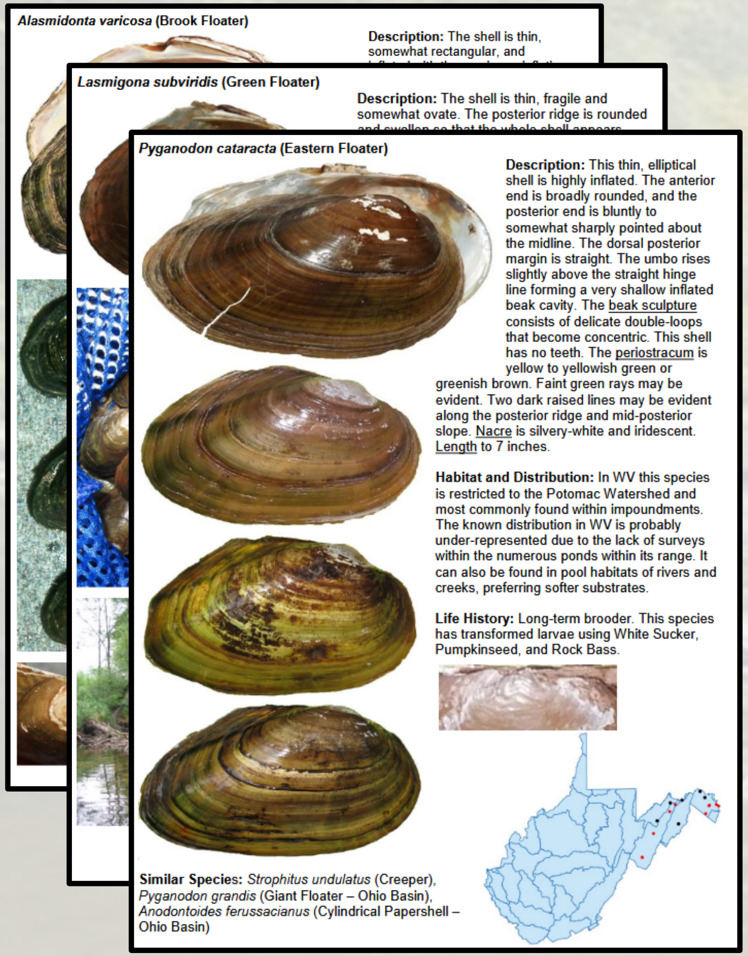
Green Floater (*L. subviridis*)

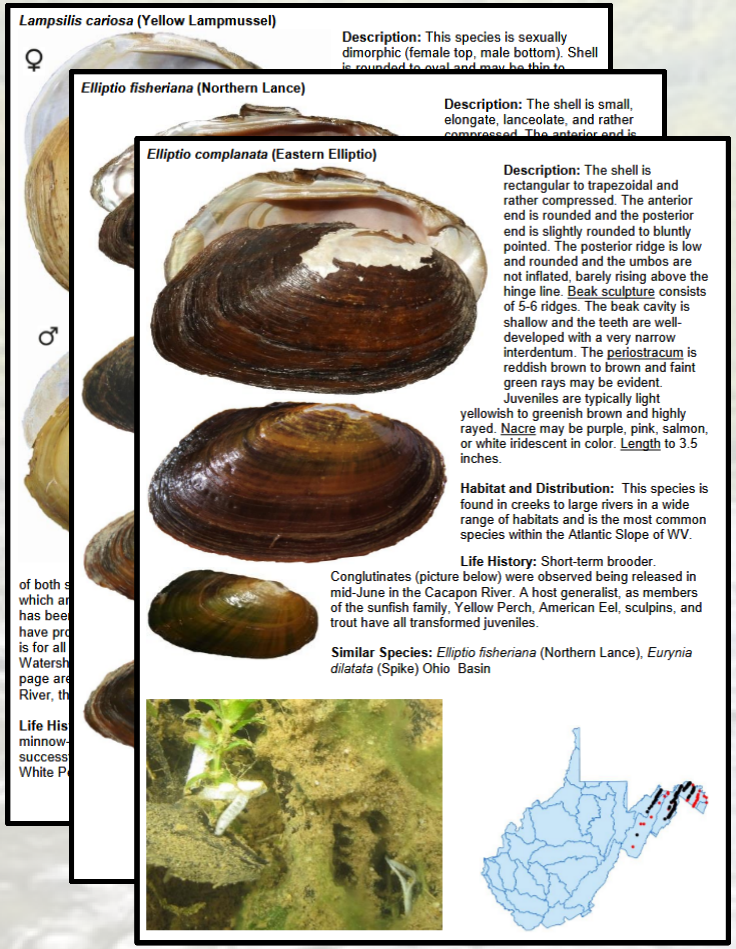


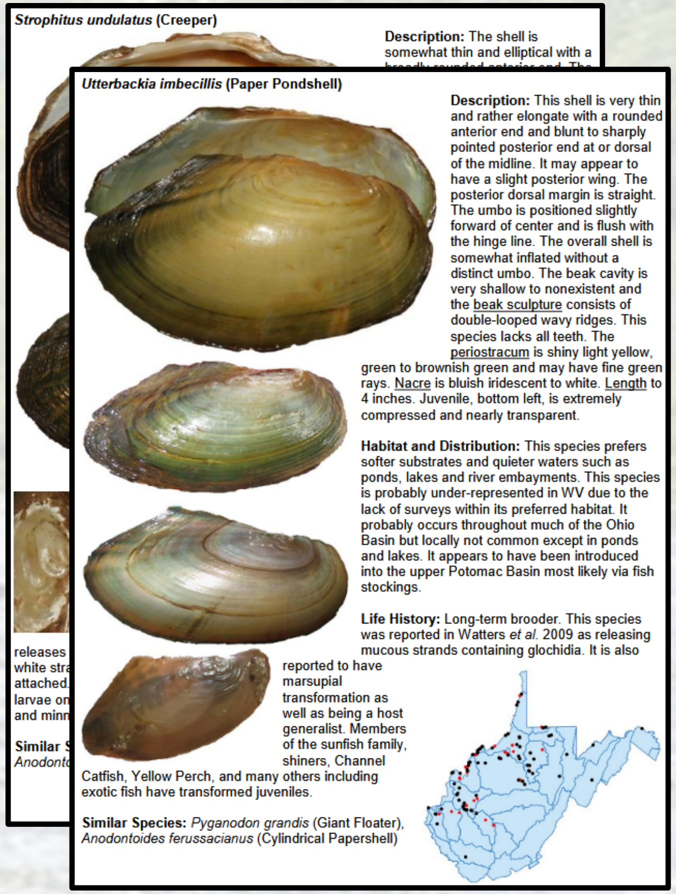
Eastern Floater (*P. cataracta*)

Modern vs Historical Distributions: Freshwater Mussels

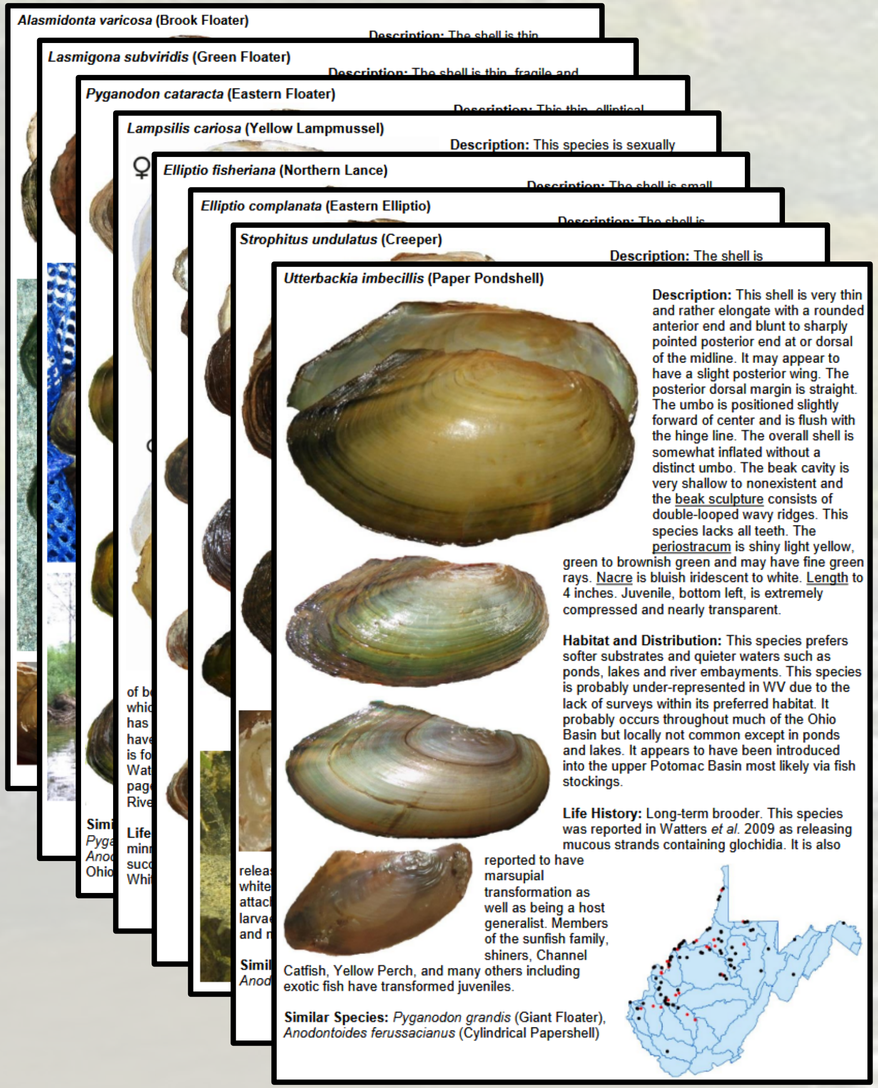
Freshwater Mussels of West Virginia
(Clayton, 2023)







Modern vs Historical Distributions: Freshwater Mussels


Alasmidonta varicosa (Brook Floater)

Lasmigona subviridis (Green Floater)

Pyganodon cataracta (Eastern Floater)

Lampsilis cariosa (Yellow Lampmussel)

Elliptio fisheriana (Northern Lance)

Elliptio complanata (Eastern Elliptio)

Strophitus undulatus (Creeper)

Utterbackia imbecillis (Paper Pondshell)

Description: This shell is very thin and rather elongate with a rounded anterior end and blunt to sharply pointed posterior end at or dorsal of the midline. It may appear to have a slight posterior wing. The posterior dorsal margin is straight. The umbo is positioned slightly forward of center and is flush with the hinge line. The overall shell is somewhat inflated without a distinct umbo. The beak cavity is very shallow to nonexistent and the beak sculpture consists of double-looped wavy ridges. This species lacks all teeth. The periostracum is shiny light yellow, green to brownish green and may have fine green rays. Nacre is bluish iridescent to white. Length to 4 inches. Juvenile, bottom left, is extremely compressed and nearly transparent.

Habitat and Distribution: This species prefers softer substrates and quieter waters such as ponds, lakes and river embayments. This species is probably under-represented in WV due to the lack of surveys within its preferred habitat. It probably occurs throughout much of the Ohio Basin but locally not common except in ponds and lakes. It appears to have been introduced into the upper Potomac Basin most likely via fish stockings.

Life History: Long-term brooder. This species was reported in Watters *et al.* 2009 as releasing mucous strands containing glochidia. It is also reported to have marsupial transformation as well as being a host generalist. Members of the sunfish family, shiners, Channel Catfish, Yellow Perch, and many others including exotic fish have transformed juveniles.

Similar Species: *Pyganodon grandis* (Giant Floater), *Anodontoides ferussacianus* (Cylindrical Papershell)

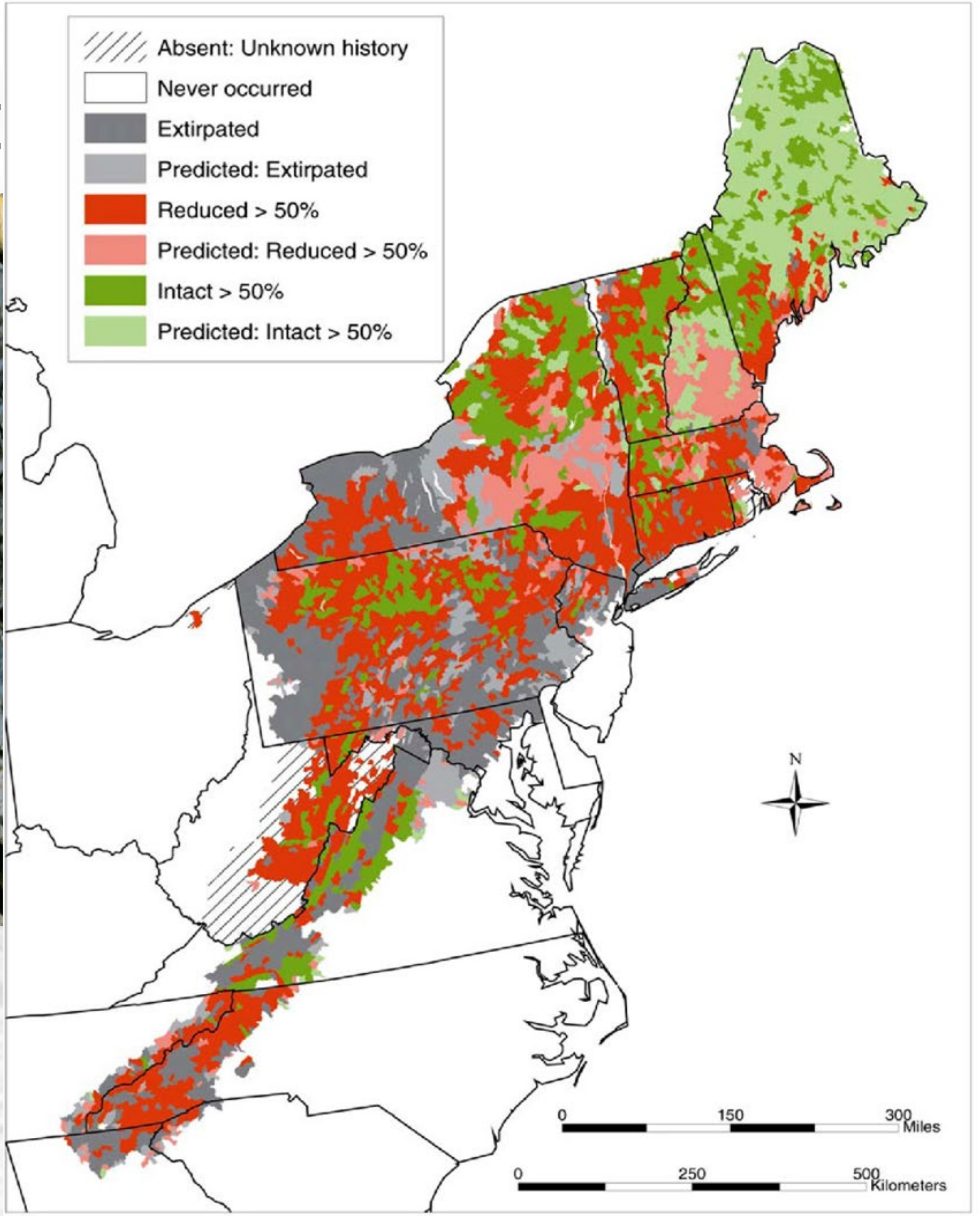
- Ortmann, A.E. **1912**. Lampsilis ventricosa (Barnes) in the Upper Potomac drainage. The Nautilus 26(4):51-54.
- Ortmann, A.E. **1913**. The Alleghenian Divide, and its influence on the freshwater fauna. Proceedings of the American Philosophical Society 52(210):287-396, pls. 12-14.
- Marshall, W.A. **1917**. Lampsilis ventricosa cohongoronta in the Potomac River. The Nautilus 31 (2):40-41.
- Marshall, W.B. **1918**. Lampsilis ventricosa cohongoronta in the Potomac River valley. The Nautilus 32(2):51-53.
- Ortmann, A.E. **1919**. A monograph on the naiades of Pennsylvania. Part 3. Systematic account of the genera and species. Memoirs of the Carnegie Museum 8(1):xiv + 384 pp. 21 Pis., 34 Figs.
- Ortmann, A.E. **1920**. Correlation of shape and station in freshwater mussels. Proceedings of the American Philosophical Society 59(4):269-312. 1 Map.
- Marshall, W.B. **1930**. Lampsilis ventricosa cohongoronta in the Potomac River. The Nautilus 44(1):19-21.
- Marshall, W.B. **1930**. Mollusks from below Conowingo Dam, Maryland. The Nautilus 43:87-88

Freshwater Mussels were extirpated from most of the North Branch watershed before taxonomists could record them.

Modern vs Historical Distributions: Freshwater Mussels

- Survey the Upper Potomac basin to establish modern distribution records
- Deploy mussel silos in the upper Potomac basin to determine habitat suitability of the mainstem and significant tributaries
- There appears to be a westward expansion towards the North Branch

Modern vs Historical Distribut



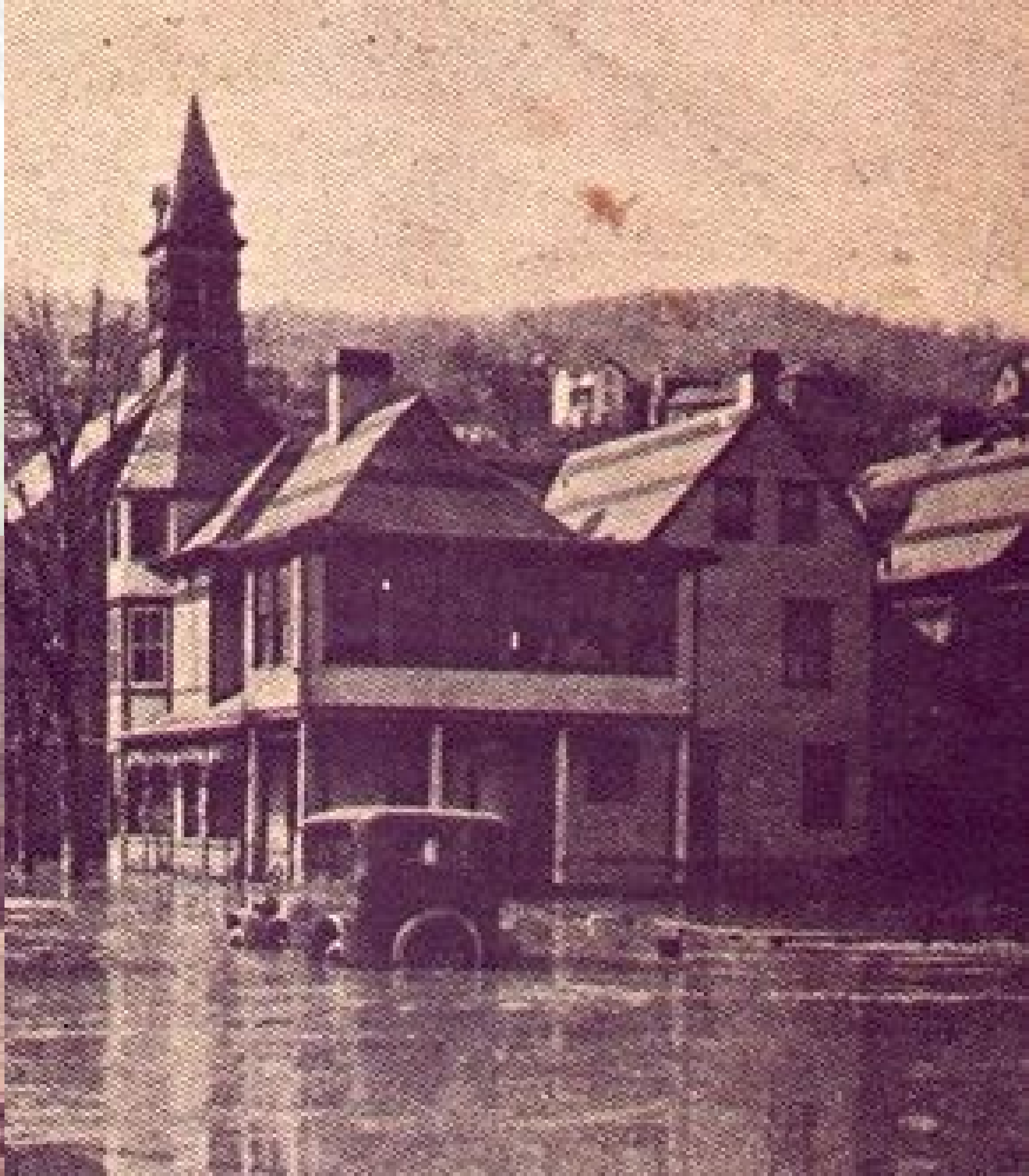
- **<u>Brook Trout</u>**: Potomac basin's only native salmonid.

1911 – Fishery surveys of upper Potomac and tributaries from 1898-1911 find 84 species of fish

EBTJV 2025

"Another nor'easter struck on the 21st, this time dumping 15 to 20 inches of snow locally. The month's heavy snow left many roads blocked, and winds produced high drifts in some cases 15 to 20 feet high. There was 3 to 4 feet of snow covering the mountains. Then, on March 28, the temperature topped 70 degrees at Cumberland."

Luke, MD (1924)



A Resource Exploited River

ICPRB Created (1940)

UPRC Created (1944)

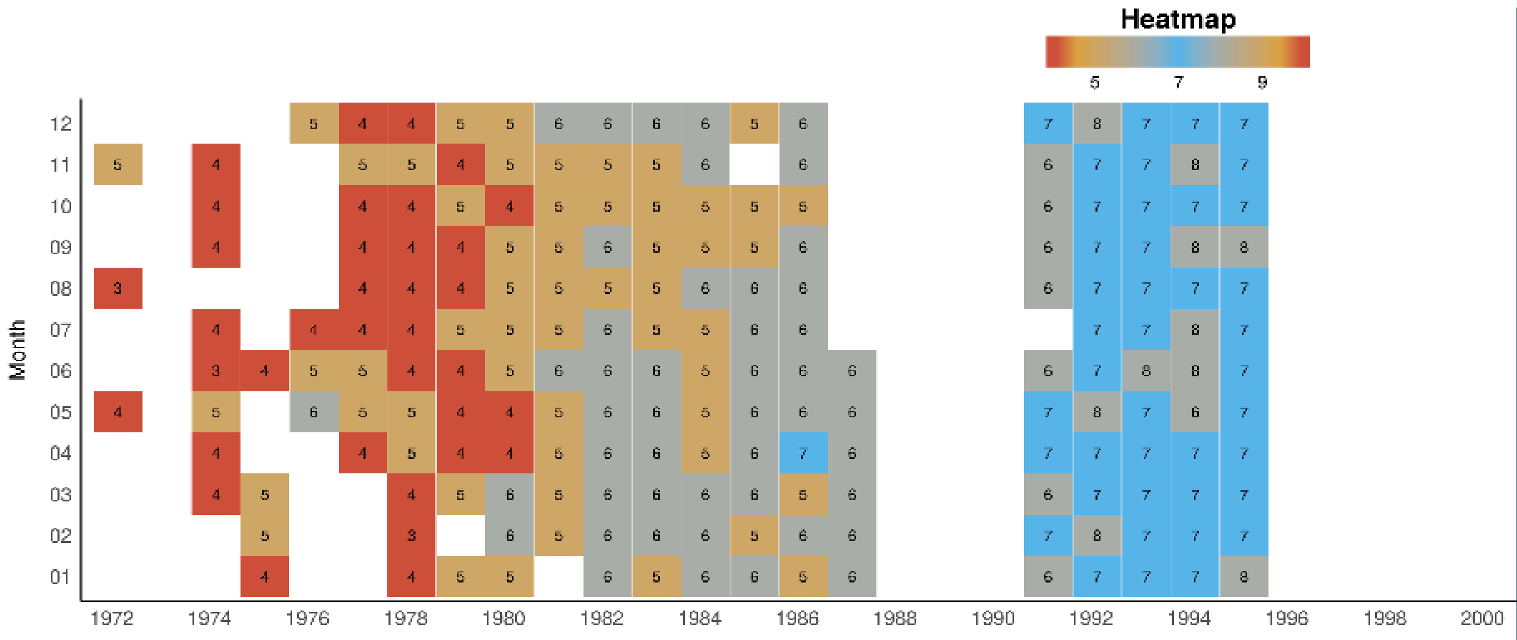
Savage River Reservoir (1952)

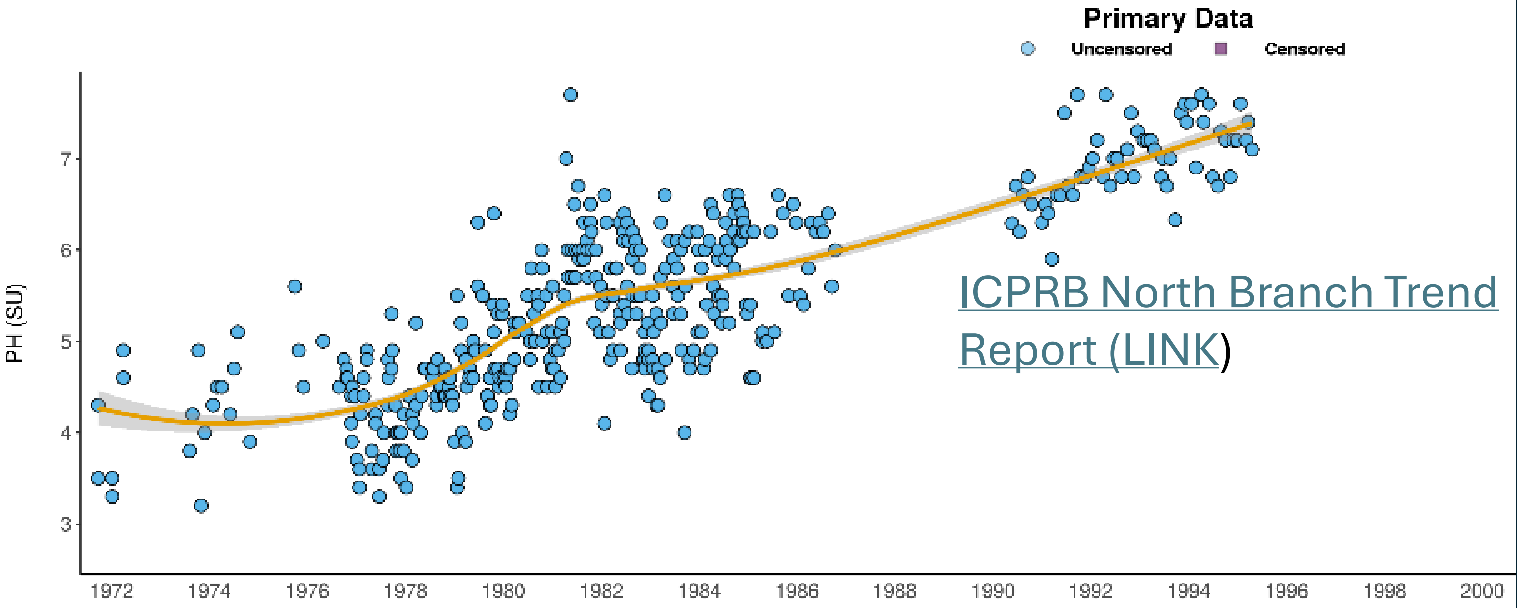
Jennings Randolph construction begins

1924 – 1965	1967	1970

NBPR consistently below pH 4.5, as low as pH 2.3

Dead River





ICPRB North Branch Trend Report (LINK)

1990

Comprehensive evaluation of JRL tributaries

Pre-doser baseline

Smallmouth Bass reintroduction

1992

First Doser installed at Kitzmiller (above JRL)

1992-1999

North Branch Task Force created

5 Dosers are installed and functionally operational

By 1999 upper NBR sites meet Maryland State Standards

2000-2009

3 additional dosers are installed

10 year review of Task Force objectives and “vision” goals redefined

2010-2025

Coordination between USACE and regional stakeholders considers biologically significant time periods to improve the tailwater

2019: Luke Paper mill shuts down after 131 years

Biological and temperature monitoring observe cold water conditions beyond the historic extent

Future

Define a new cooperative vision for the watershed

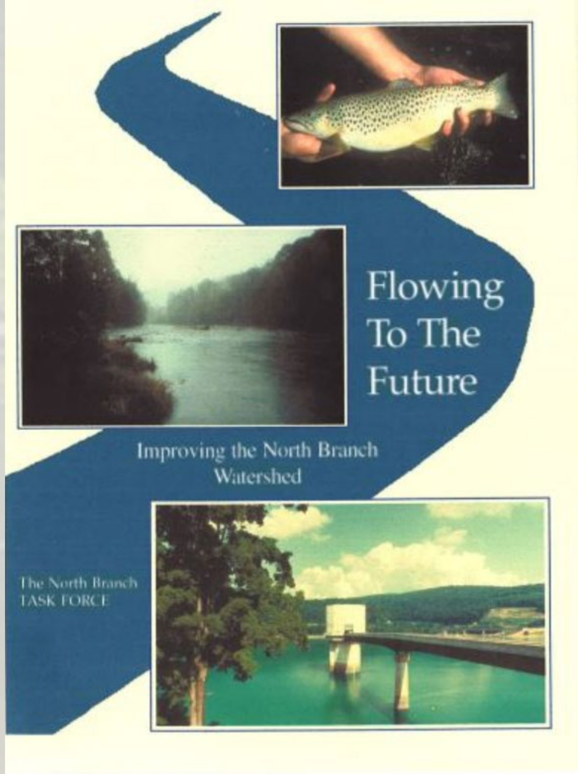
Recovery Continues

Addressing Data Gaps and Needs

North Branch Potomac River Tailwater JRL to Cumberland

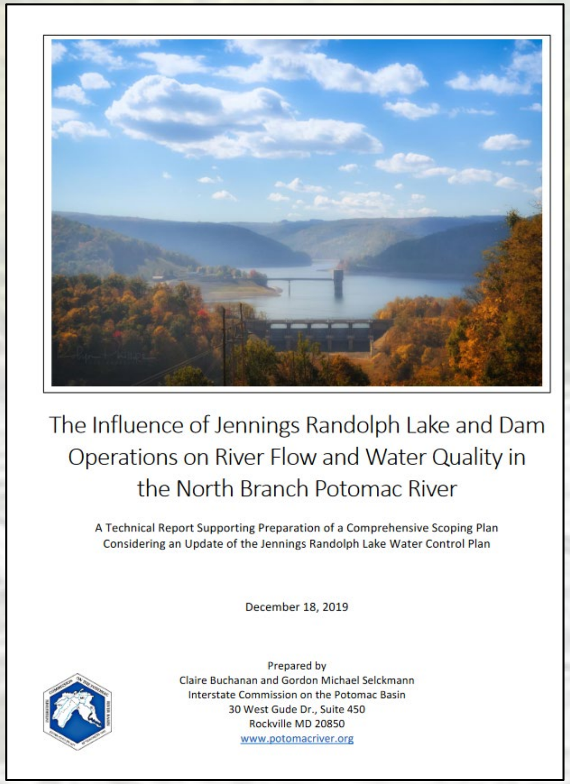
North Branch Vision Plan

North Branch Advisory Committee



Link: North Branch Advisory Committee (NBAC)

Considerations for updating the USACE Water Control Plan



The Influence of Jennings Randolph Lake and Dam Operations on River Flow and Water Quality in the North Branch Potomac River

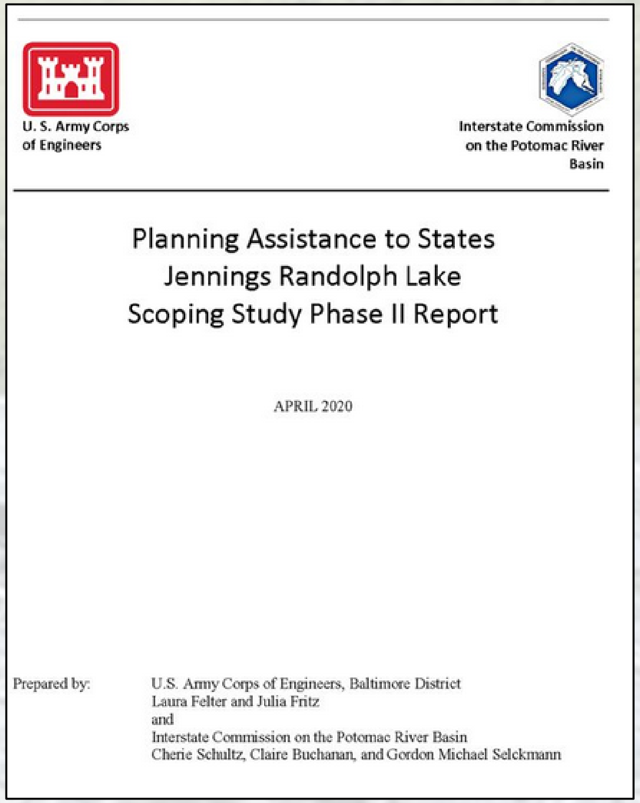
A Technical Report Supporting Preparation of a Comprehensive Scoping Plan Considering an Update of the Jennings Randolph Lake Water Control Plan

December 18, 2019

Prepared by
Claire Buchanan and Gordon Michael Selckmann
Interstate Commission on the Potomac Basin
30 West Gude Dr., Suite 450
Rockville MD 20850
www.potomacriver.org

Link: 2019 ICPRB Report

USACE: Scoping Study



U. S. Army Corps of Engineers

Interstate Commission on the Potomac River Basin

Planning Assistance to States
Jennings Randolph Lake
Scoping Study Phase II Report

APRIL 2020

Prepared by: U.S. Army Corps of Engineers, Baltimore District
Laura Felter and Julia Fritz
and
Interstate Commission on the Potomac River Basin
Cherie Schultz, Claire Buchanan, and Gordon Michael Selckmann

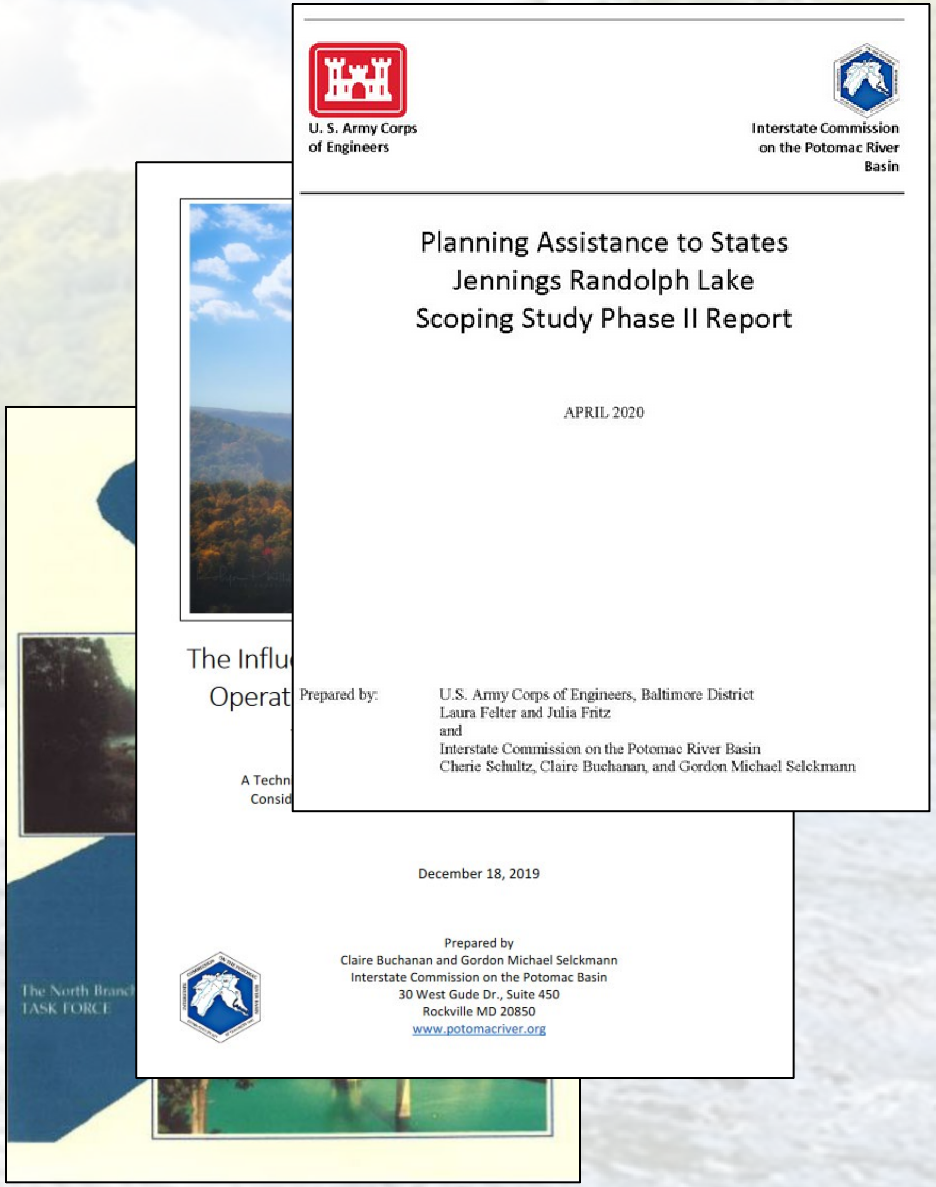
Link: 2020 USACE Report

Addressing Data Gaps and Needs

North Branch Potomac River Tailwater JRL to Cumberland

Questions generated from these projects/workgroups:

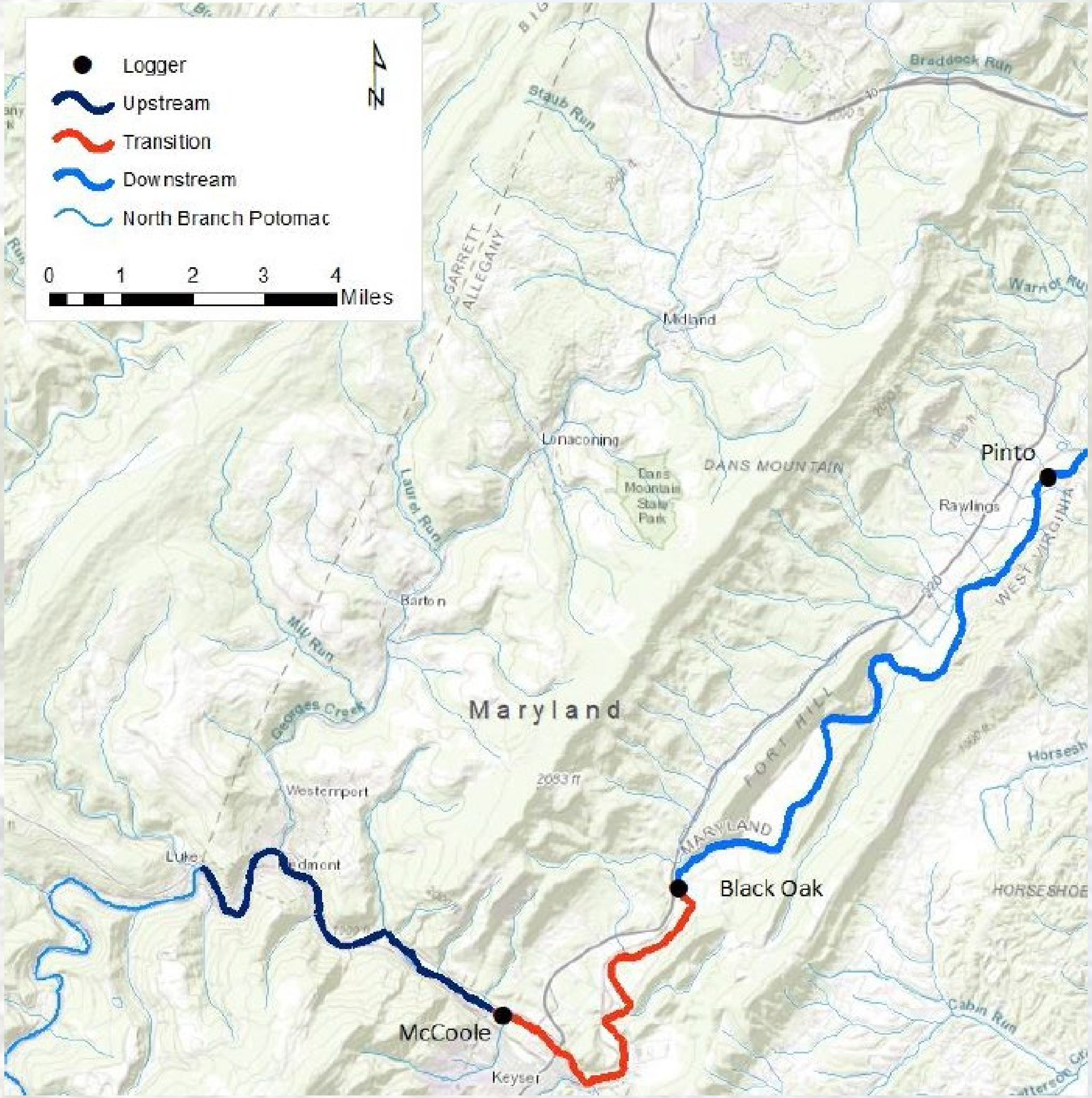
1. What is the downstream extent and influence of reservoir operations?
2. Are tailwater biological communities' behaviors and distributions linked to reservoir operations?
3. Are tailwater biological communities capable of responding to short term (AVFs, whitewater events, thermal releases) and long term (seasonal) stressors?



<u>Data Gap/Need:</u> Temperature Monitoring
<u>Action:</u> Temperature Logging Network (2016-current)

Data Gap/Need: Biological Response to Reservoir Actions Action: Telemetry Survey (2023)

- Luke to McCoole
  - Thermally stable reach
  - Temperatures remain in metabolic “goldilocks zone” for cold water species
  - **Hypothesis: Local movement**
- McCoole to Black Oak
  - “Transition” zone from cool to warm
  - Historically the end of USACE effective reach
  - **Hypothesis: Reservoir release/temperature linked movement. Temperature avoidance.**
- Black Oak to Pinto
  - Warm water reach
  - **Hypothesis: Reservoir release/temperature linked movement. Temperature avoidance.**



Data Gap: Biological Response to Reservoir Actions
Action: Telemetry Survey

- Deploy continuous water and air temperature loggers

- 25 naturalized Rainbow Trout (12”-18”) received radio transponders and were released at 11 stations

- 5 - Upper Section
- 9 - Middle Section
- 11 - Lower Section

Null Hypothesis: Rainbow trout will not respond to thermal or flow linked stressors and will remain proximal to their point of release.

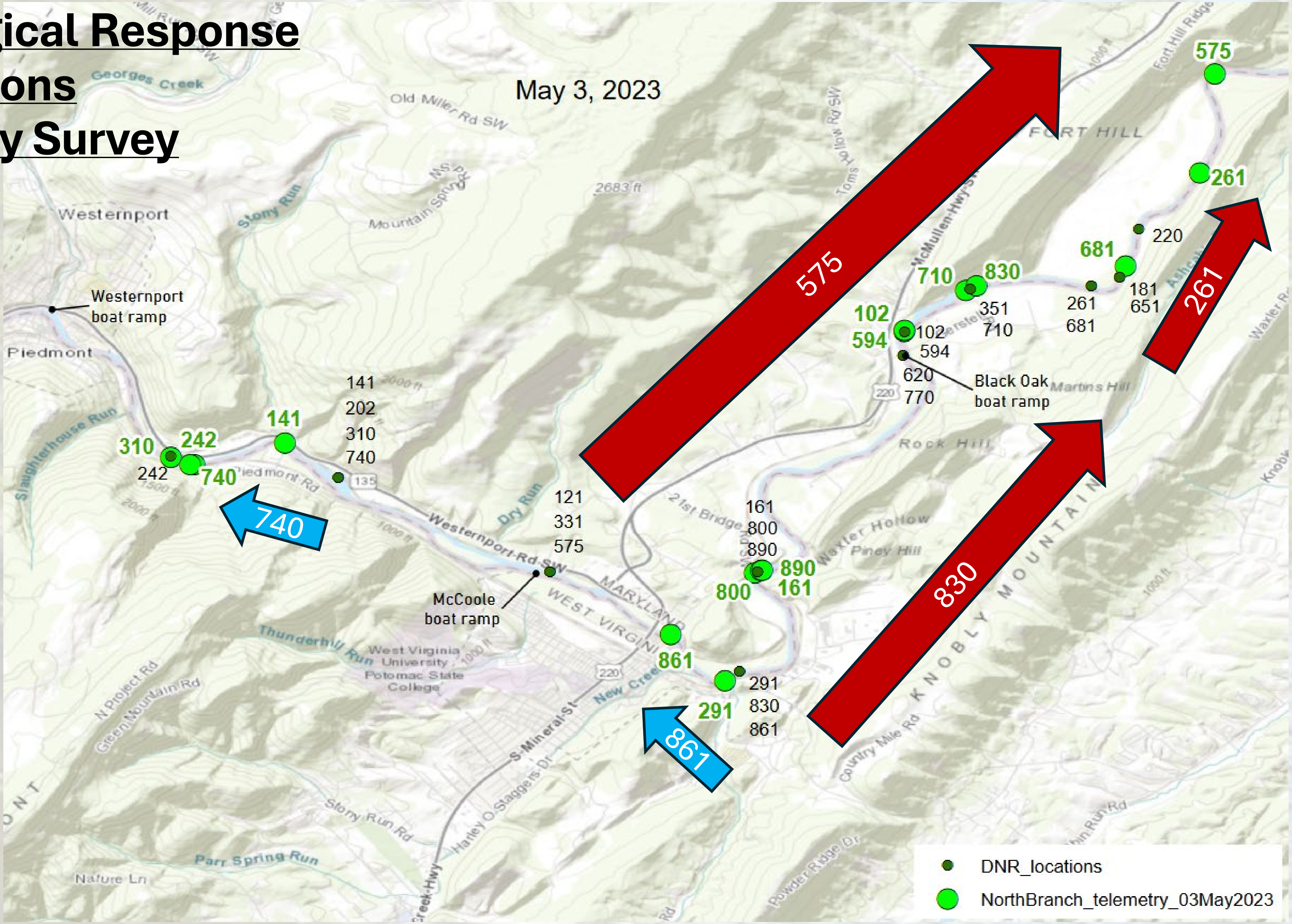
Matt Sell (MDDNR) and John Mullican (MDDNR) implanting transponders following the electroshocking cataraft.

Data Gap: Biological Response to Reservoir Actions
Action: Telemetry Survey

- Movement was observed in both upstream and downstream directions

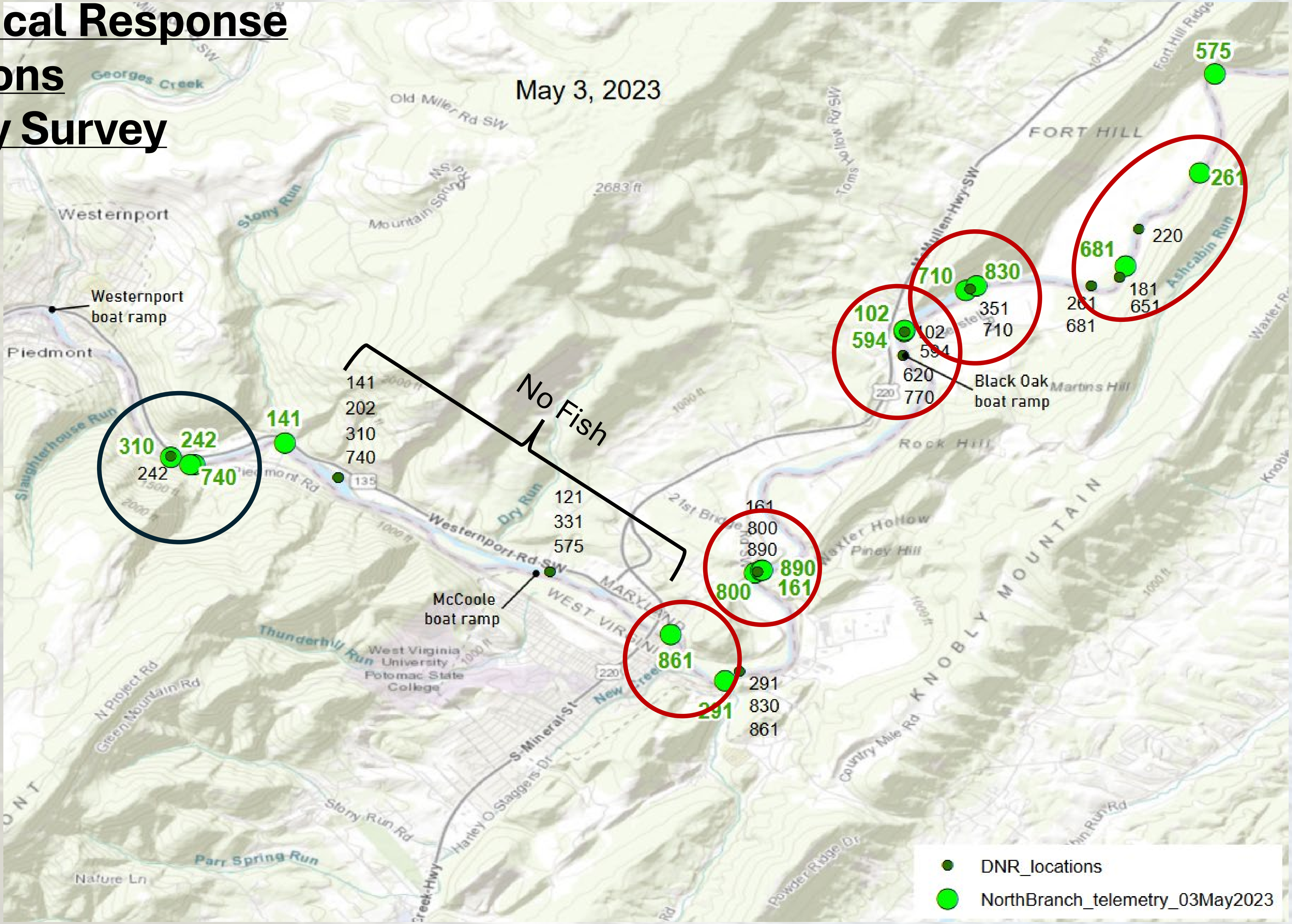
- Large downstream movement to new reaches.

- Small movements observed between nearby habitats.



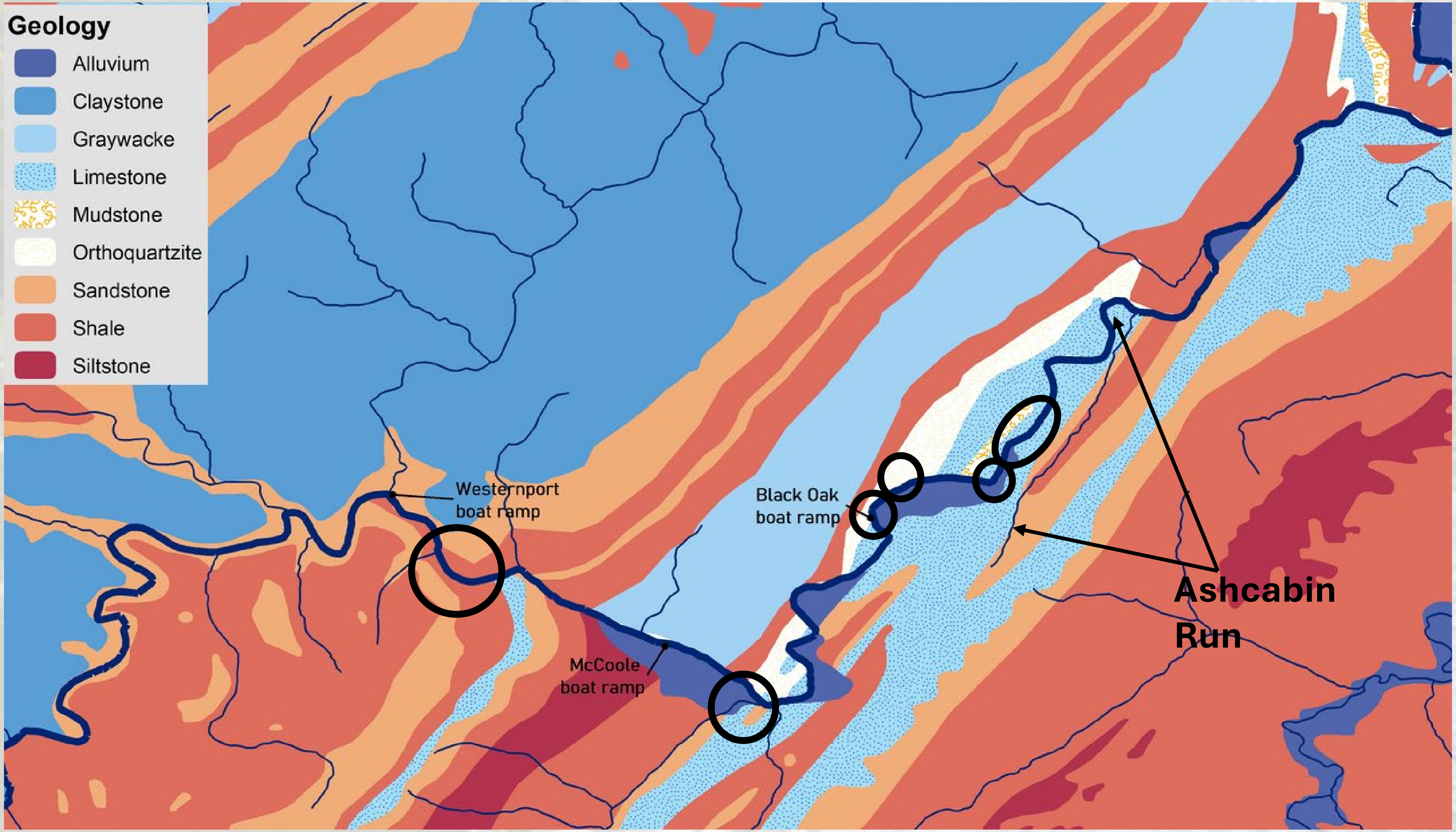
Data Gap: Biological Response to Reservoir Actions
Action: Telemetry Survey

- Fish appeared to cluster at predictable locations in the summer
- Clustering dispersed in autumn through early spring

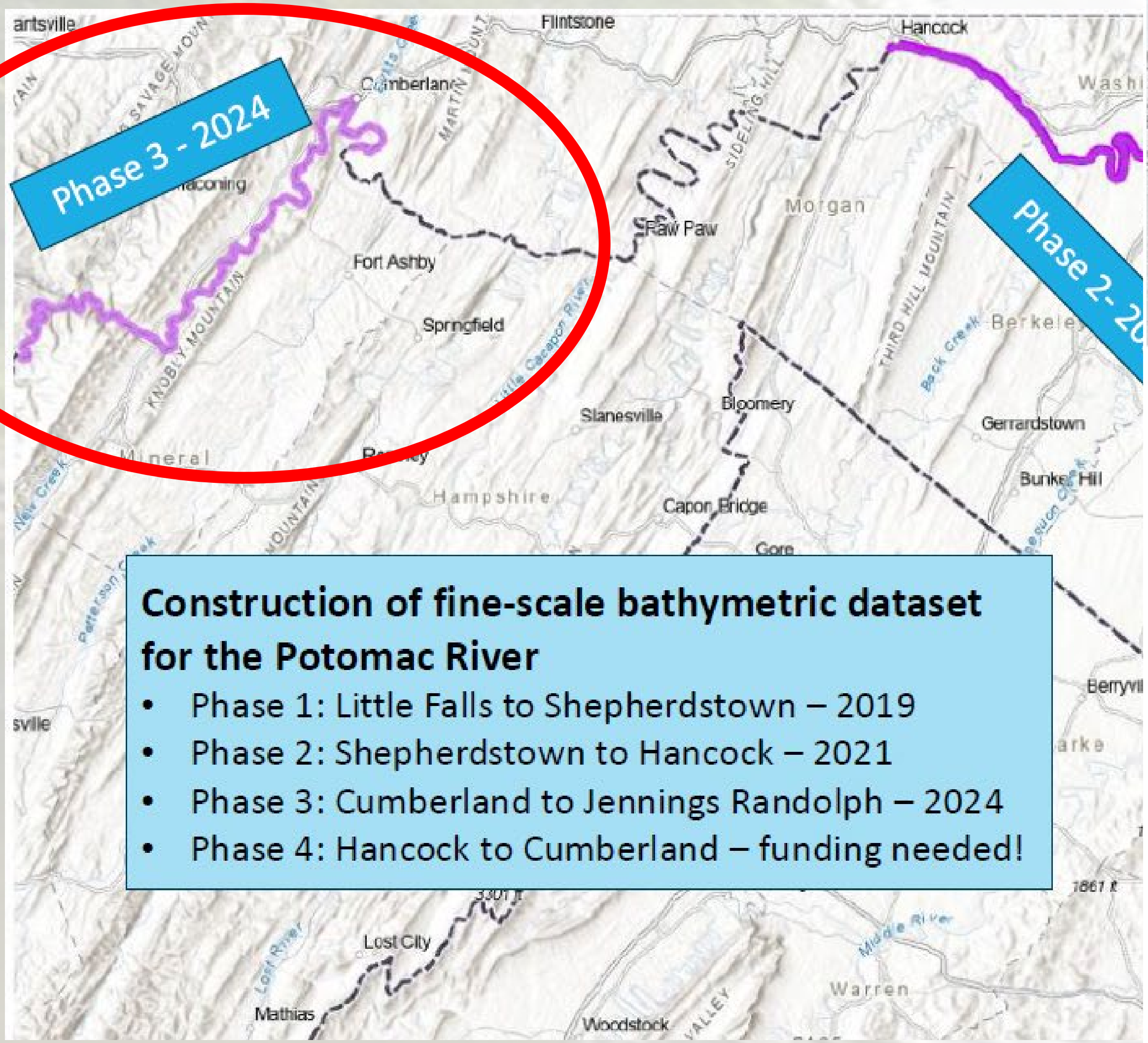


Data Gap/Needs: Biological Response to Reservoir Actions
Action: Telemetry Survey

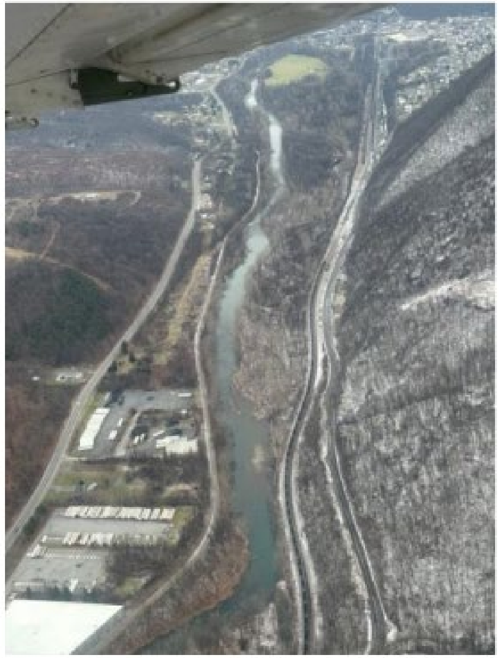
Groundwater thermal refugia?



Data Gap: River morphology and critical habitats
Action: LiDAR Survey



Construction of fine-scale bathymetric dataset for the Potomac River
- Phase 1: Little Falls to Shepherdstown – 2019
- Phase 2: Shepherdstown to Hancock – 2021
- Phase 3: Cumberland to Jennings Randolph – 2024
- Phase 4: Hancock to Cumberland – funding needed!



Data Gap/Need
data to operatic
Action: Explore

prediction sites

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  - ✓Accounts fo autocorrelati both Euclidea
  - ✓Provides ar process-base
- What is it goo
  - ✓Assess and endpoints
  - ✓Predict phy characteristic
  - ✓Support TM

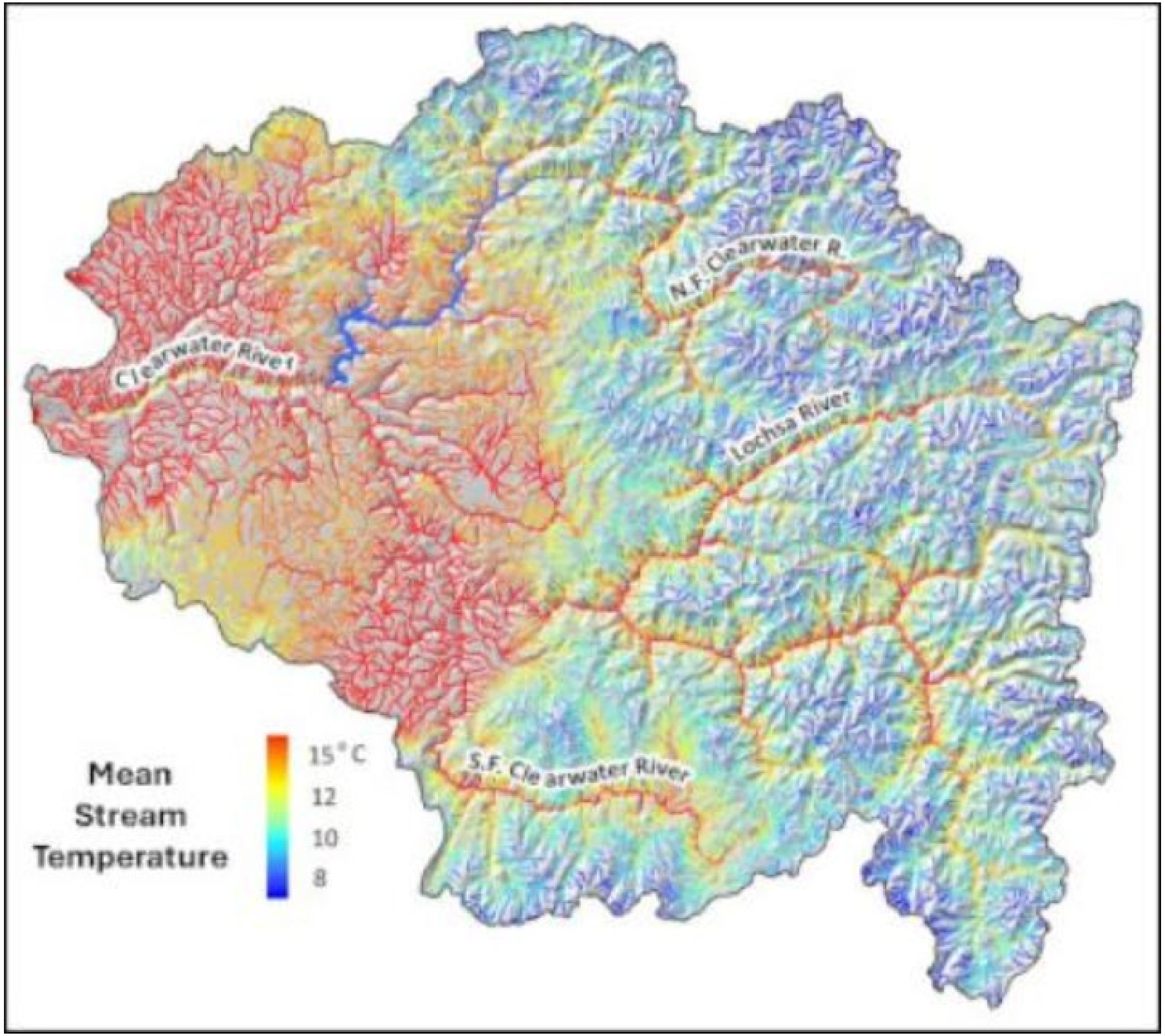
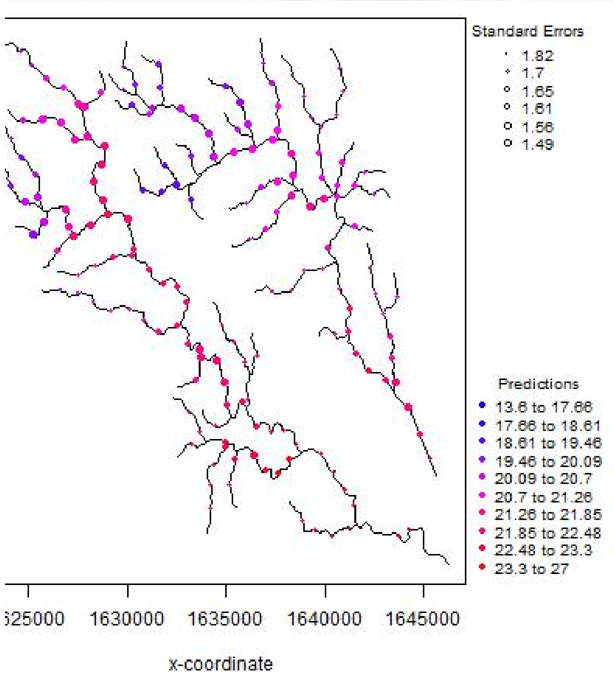
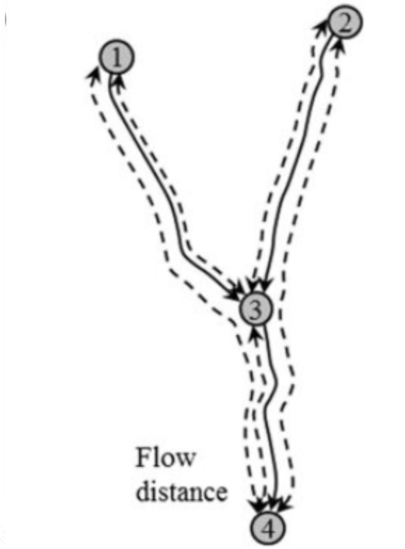


Photo Credit: *David Nagel, RMRS Boise Aquatic Sciences Lab*





al-stream-network#resources

Data Gap/Need: Biological Monitoring

Action: Respond to state's needs

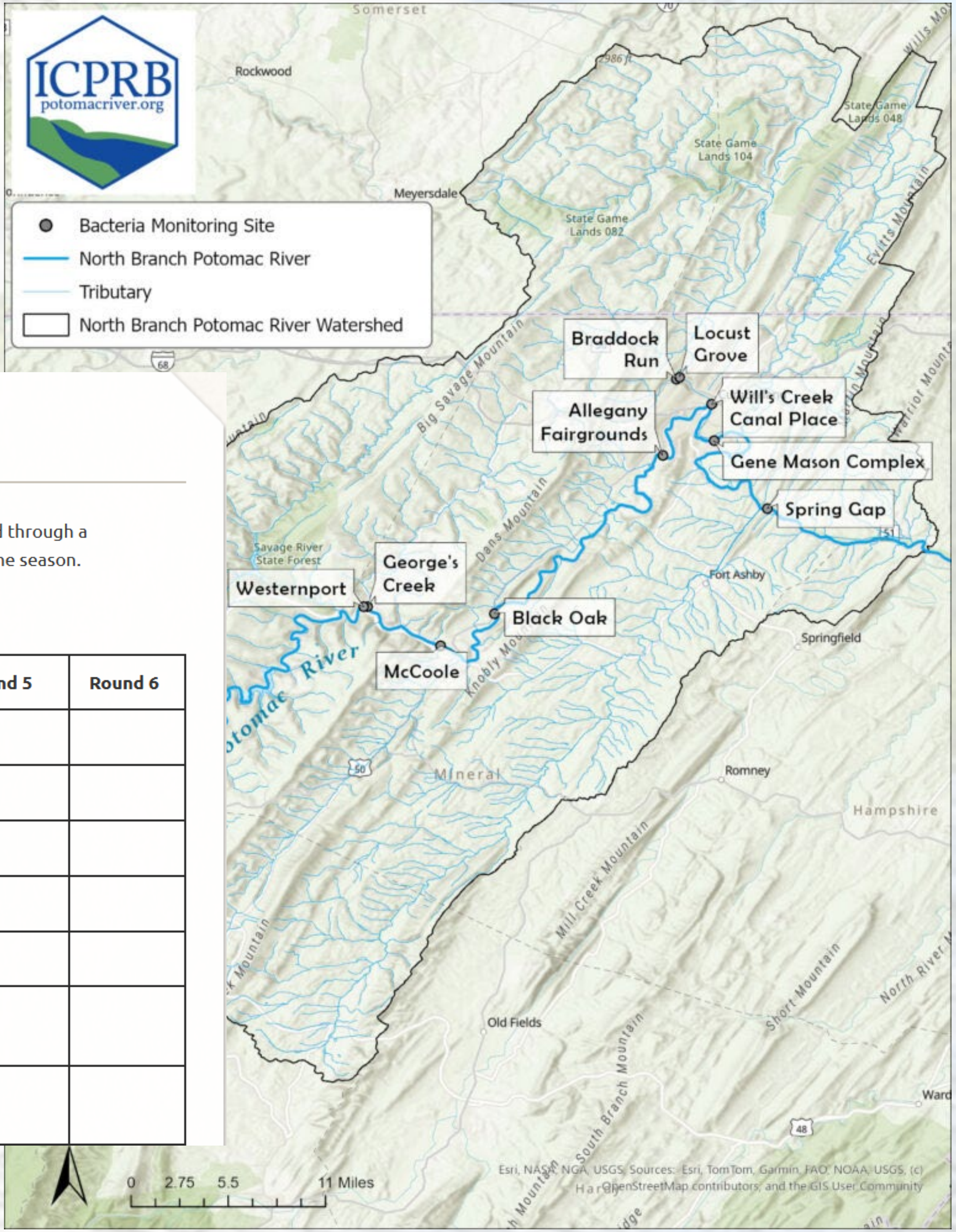
NORTH BRANCH BACTERIAL MONITORING

Interstate Commission on the Potomac River Basin

The following bacteria data is being collected during the summer of 2025 at locations along the North Branch Potomac River. It is provided through a partnership between ICPRB and the Maryland Department of Environment. ICPRB will conduct bi-weekly monitoring through the end of the season.

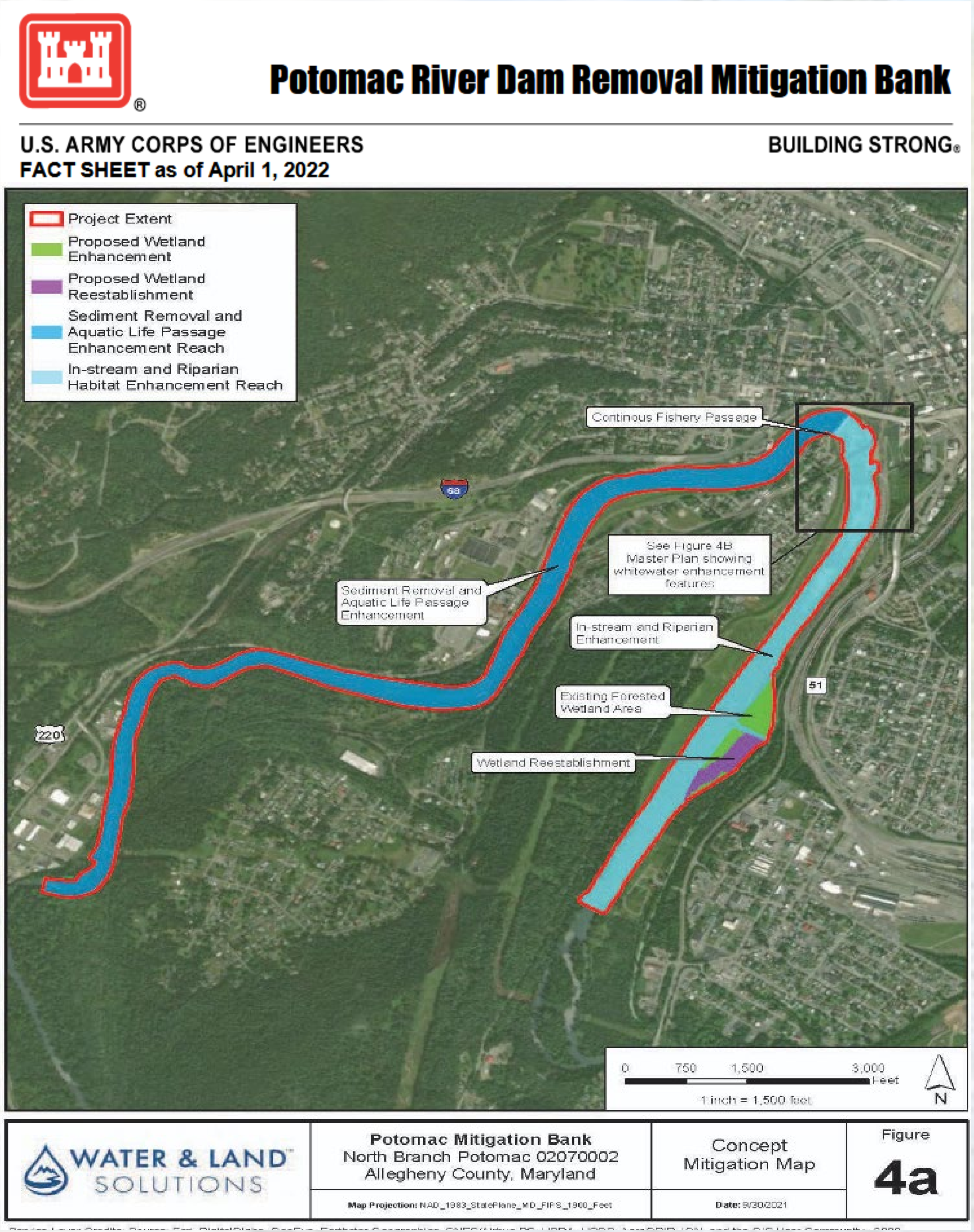
Data

Site Description	Type	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6
North Branch Potomac us stream of Westernport	Mainstem	56	1986	1300			
Mouth of Georges Creek at Boat Launch	Tributary	1300	>2419	>2419			
McCoole Boat Launch, North Branch Potomac	Mainstem	71	60	104			
Black Oak Boat Launch, North Branch Potomac	Mainstem	82	51	116			
Fairground's Boat Launch, North Branch Potomac	Mainstem	47	89	34			
Gene Mason Recreation Complex Boat Launch (Above Cumberland WWTP)	Mainstem	40	162	122			
Braddock Run at LaVale Pump Station on Old Mt. Savage Rd.	Tributary	>2419	649	1986			



Future of the North Branch Potomac

- Reservoir zone of Influence
  - Drought/Climate resiliency
  - Stressor/Crisis response
  - New stakeholders
- Fish Passage/Stream Connectivity
  - Target species resiliency
  - Define critically import locations and time periods
  - Invasive species
- Mussel re-establishment



Questions/ Comments?

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